

Glossary: Cognitive Aspects

Cognition

The mental processes used to **think, learn, remember, and understand information**.

Example: reading a book and understanding its meaning

Experiential Cognition

Fast and automatic thinking that happens **without much effort or awareness**.

Examples:

1. driving a familiar route.
2. typing on a keyboard without looking.

Reflective Cognition

Slow and careful thinking that involves **analysis, reasoning, and decision-making**.

Example: planning a project or solving a complex question

Attention

The ability to **focus on important information while ignoring distractions**.

Example: listening to a lecture while ignoring noise

Perception

The process of **interpreting information received through the senses**.

Example: recognizing a red light as a signal to stop

Memory

The ability to **store and retrieve information when needed**.

Example: remembering a friend's phone number

Recall

Retrieving information from memory **without any external help or cues**.

Example: Writing an answer in an exam without looking at notes.

Recognition

Identifying information when it is **seen or heard again**.

Example: Selecting the correct option from a multiple-choice question.

Working Memory

A short-term system that **temporarily holds and processes small amounts of information** needed for immediate tasks..

Example: Remembering a phone number just long enough to dial it.

Long-Term Memory

Memory that **stores information for a long time, even permanently**. It includes knowledge, experiences, and skills.

Example: remembering how to ride a bicycle

Encoding

The process of **converting information into a form that can be stored in memory**.

Example: taking notes to remember a topic

Context

The situation or environment in which information is learned or remembered, which can affect how well it is recalled.

Example: remembering something better in the same place you learned it

Multitasking

Performing multiple tasks at the same time, which usually **reduces focus and efficiency**.

Example: studying while checking social media

Cognitive Load

The amount of **mental effort required to complete a task**. High cognitive load makes tasks harder and more error-prone.

Example: remembering many passwords at once

Learning

The process of **gaining new knowledge or skills through study or experience**.

Example: learning a new programming language

Language Processing

The ability to **understand and use language through reading, writing, listening, and speaking**.

Example: understanding a lecture or conversation

Problem-Solving

The process of **finding solutions to challenges or difficult situations**.

Example: fixing a bug in a program

Decision-Making

The process of **choosing the best option from available alternatives**.

Example: Choosing between two job offers.

Mental Model

A person's **internal understanding of how something works**.

Example:

1. understanding how a website navigation works
2. understanding how an ATM machine works

Erroneous Mental Model

An incorrect understanding that **leads to mistakes or confusion**.

Example: thinking pressing a button multiple times makes it faster

Cognitive Framework

A set of ideas or theories used to **explain how people think and interact with systems**.

Example: using psychological theories to design user-friendly apps

Personal Information Management

The way people **organize, store, and retrieve their personal digital information**.

Example: using folders or search to find files

Multifactor Authentication (MFA)

A security method that requires **two or more types of verification to confirm identity**.

Example: password + OTP

Memory Aids

Tools or techniques that **help people remember information more easily**.

Example: reminders or notes